

Methodology

[melbrt/bayesian-analysis-film-adaptations](https://melbrt.com/bayesian-analysis-film-adaptations)

1. Research Question

The objective of this project is to determine whether films released after 2010 are more likely to be adaptations of pre-existing works than films released before 2010. More specifically, we seek to estimate and compare the probability that a film is an adaptation during two different historical periods using Bayesian statistical inference.

The research question is therefore:

Has the probability that a film is an adaptation increased after 2010 compared with before 2010?

2. Hypotheses

Let:

- θ_{before} = probability that a film released before 2010 is an adaptation
- θ_{after} = probability that a film released after 2010 is an adaptation

The hypotheses are:

Null Hypothesis (H0)

$$\theta_{\text{after}} \leq \theta_{\text{before}}$$

The probability that a film is an adaptation has not increased after 2010.

Alternative Hypothesis (H1)

$$\theta_{\text{after}} > \theta_{\text{before}}$$

The probability that a film is an adaptation is higher after 2010.

The purpose of the Bayesian analysis is to estimate these probabilities and evaluate the evidence supporting H1.

3. Data Collection

The dataset was obtained from Wikidata using a SPARQL query. Wikidata is a collaborative knowledge base that stores structured information about millions of entities, including films and their associated metadata.

Only entities identified as films were selected using the Wikidata property:

- P31 = instance of film (Q11424)

For each film, the query retrieved the following variables:

- film identifier
- film title
- release year
- box office revenue (when available)
- adaptation status
- time period

The dataset therefore contains observations such as:

	film	filmLabel	year	boxOffice	isAdaptation	period
0	http://www.wikidata.org/entity/Q553901	Uncle Boonmee Who Can Recall His Past Lives	2010	184292.0	0	After 2010

The adaptation variable is based on Wikidata property P144 ("based on").

A binary variable called *isAdaptation* was created directly within the SPARQL query:

```

BIND(
  IF(EXISTS { ?film wdt:P144 ?x }, 1, 0)
  AS ?isAdaptation
)
```

Films associated with at least one source work receive the value:

- 1 = adaptation

Films without this property receive:

- 0 = non-adaptation

A second variable called *period* was also generated directly in the query using the film's release year:

```

BIND(
  IF(YEAR(?year) < 2010,
    "Before 2010",
    "After 2010")
)
```

AS ?period

This variable divides films into two groups corresponding to the research question and allows direct comparison of adaptation probabilities before and after 2010.

Although box office information was retrieved when available, it was not used in the Bayesian model itself. The statistical analysis focuses exclusively on adaptation status and release period.

4. Data Preparation

The dataset extracted from Wikidata was imported into Python using the pandas library.

Before conducting the statistical analysis, the data were inspected and cleaned. Duplicate entries were removed when necessary to ensure that each film was counted only once.

The dataset already contains a variable called *period*, which was created directly within the SPARQL query using the release year of each film. This variable classifies films into two categories:

- **Before 2010**
- **After 2010**

The total number of films in each period was then calculated:

These quantities represent the sample sizes used in the Bayesian analysis and correspond to the total number of observations available in each period.

5. Descriptive Statistics

Before estimating Bayesian posterior distributions, descriptive statistics were computed for each period.

The number of films (n), the number of adaptations (k), and the observed adaptation rate (k/n) were calculated separately for films released before and after 2010. These descriptive statistics provide an initial summary of the data and serve as the basis for the Bayesian analysis presented in the following sections. The resulting summary table is presented in the report.

6. Bayesian Statistical Model

6.1 Bernoulli Model

The objective of this study is to estimate the probability that a film is an adaptation and to compare this probability before and after 2010. For each film, the adaptation status is represented by a binary variable. A film can either be classified as an adaptation or as a non-adaptation. Consequently, the outcome naturally follows a Bernoulli distribution.

Let:

- $Y = 1$ if the film is an adaptation;
- $Y = 0$ otherwise.

The adaptation status of each film is therefore modeled as:

$$Y \sim \text{Bernoulli}(\theta)$$

where θ represents the probability that a randomly selected film is an adaptation.

Because the response variable is binary, the Bernoulli model provides an appropriate likelihood function for the analysis.

6.2 Prior Distribution

A Bayesian analysis requires the specification of a prior distribution for the unknown parameter θ . Since no strong prior assumptions were made regarding adaptation probabilities, a non-informative Beta prior distribution was selected:

$$\theta \sim \text{Beta}(1,1)$$

The Beta(1,1) distribution corresponds to a uniform distribution over the interval [0,1]. This choice assigns equal prior plausibility to all possible adaptation probabilities and allows the data to drive the inference.

In the implementation, the prior parameters were defined as:

$$\alpha_0 = 1$$

$$\beta_0 = 1$$

This prior was applied independently to both periods under investigation.

6.3 Posterior Distribution

One of the advantages of the Beta distribution is that it is conjugate to the Bernoulli likelihood. As a result, posterior distributions can be obtained analytically without requiring complex numerical estimation procedures.

For a sample containing:

- n films;
- k adaptations;

the posterior distribution is given by:

$$\theta \mid \text{data} \sim \text{Beta}(1+k, 1+n-k)$$

Posterior distributions were estimated separately for films released before 2010 and films released after 2010.

The posterior parameters were computed as follows:

$$a_{\text{before}} = \alpha_0 + k_{\text{before}}$$

$$b_{\text{before}} = \beta_0 + (n_{\text{before}} - k_{\text{before}})$$

$$a_{\text{after}} = \alpha_0 + k_{\text{after}}$$

$$b_{\text{after}} = \beta_0 + (n_{\text{after}} - k_{\text{after}})$$

This procedure produced two posterior distributions:

- θ_{before} : probability that a film released before 2010 is an adaptation;
- θ_{after} : probability that a film released after 2010 is an adaptation.

These posterior distributions summarize all available information regarding adaptation probabilities after observing the data.

7. Posterior Summaries

7.1 Posterior Means

Posterior means were calculated in order to obtain point estimates of adaptation probabilities for both periods.

The following computations were performed:

$$\text{mean_before} = \text{beta.mean}(a_{\text{before}}, b_{\text{before}})$$

$$\text{mean_after} = \text{beta.mean}(a_{\text{after}}, b_{\text{after}})$$

The posterior mean corresponds to the expected value of the adaptation probability after taking both prior information and observed data into account.

These values provide an estimate of the probability that a film is an adaptation before and after 2010.

7.2 Credible Intervals

In addition to point estimates, 95% credible intervals were computed for both posterior distributions:

$$\text{ci_before} = \text{beta.interval}(0.95, a_{\text{before}}, b_{\text{before}})$$

$$\text{ci_after} = \text{beta.interval}(0.95, a_{\text{after}}, b_{\text{after}})$$

Credible intervals quantify the uncertainty associated with the estimated probabilities.

Unlike frequentist confidence intervals, Bayesian credible intervals have a direct probabilistic interpretation. A 95% credible interval indicates that, given the model and the observed data, there is a 95% probability that the true parameter value lies within the interval.

These intervals provide important information regarding the precision of the estimates and the degree of overlap between adaptation probabilities before and after 2010.

8. Bayesian Comparison Between Periods

8.1 Monte Carlo Sampling

Although posterior means and credible intervals provide useful summaries, the central objective of the study is to compare adaptation probabilities across periods.

To achieve this, Monte Carlo sampling was performed using the posterior distributions obtained previously.

A total of 100,000 random samples were generated from each posterior distribution:

```
samples_before = np.random.beta(a_before, b_before, 100000)
```

```
samples_after = np.random.beta(a_after, b_after, 100000)
```

These samples approximate the full posterior distributions of adaptation probabilities and allow direct probabilistic comparisons between periods.

8.2 Posterior Difference

For each simulation, the difference between adaptation probabilities was computed:

$$\Delta = \theta_{\text{after}} - \theta_{\text{before}}$$

using:

```
diff = samples_after - samples_before
```

The resulting distribution of differences provides information about both the magnitude and direction of the change in adaptation probabilities.

Positive values of Δ indicate that adaptation probabilities are higher after 2010, whereas negative values indicate the opposite.

8.3 Probability of Increase

The primary quantity of interest is the posterior probability that adaptation rates increased after 2010.

This probability was estimated by calculating the proportion of simulations for which:

$$\theta_{\text{after}} > \theta_{\text{before}}$$

The computation was implemented as:

```
prob = np.mean(diff > 0)
```

This quantity directly answers the research question by estimating the probability that adaptation rates are higher after 2010 than before 2010.

Unlike traditional significance testing, Bayesian inference provides a direct probability statement regarding the hypothesis of interest.

8.4 Credible Interval for the Difference

To further assess the magnitude of the difference, a 95% credible interval was computed for the posterior distribution of Δ :

```
ci_diff = np.quantile(diff, [0.025, 0.975])
```

This interval provides information regarding the uncertainty surrounding the estimated difference between adaptation probabilities.

If the interval lies entirely above zero, this constitutes strong evidence that adaptation rates increased after 2010.